OS Lab Exam Prep: Miscellaneous Problems

TASK 01:

You are a system programmer asked to create a program that demonstrates process creation and command execution in Linux. Your program should perform the following tasks:

- The parent process should display the list of all files and directories in the current working directory.
- The child process should display the absolute path of the current working directory.

TASK 02:

As part of a Linux process management exercise, you need to create a program that demonstrates synchronization between parent and child processes. Your program should:

- The child process should create a new directory named TestDir.
- The parent process should wait for the child to finish, then list all files and directories in the current location to verify that TestDir has been created.

TASK 03:

You are a system programmer asked to demonstrate inter-process communication in Linux using a named pipe (FIFO).

- Create a named pipe called myfifo.
- The child process should send a message to the parent process through this named pipe.
- The parent process should read the message from the named pipe and display it on the screen.

TASK 04:

Explain the difference between the kill and pkill commands in Linux. Give a scenario where pkill is more convenient than kill.

TASK 05:

What is the difference between chmod and chown commands in Linux? Provide an example scenario where both are used together.

TASK 06:

Compare the ps and top commands in Linux. When would you use top instead of ps?

TASK 07:

Explain the difference between > and >> operators in Linux. Give a scenario showing how overwriting and appending differ.

TASK 08:

- Write a shell script that defines a function called search_word.
- The function should take two arguments: a word and a filename.
- Inside the function, search for the word in the given file and display all matching lines along with their line numbers.
- If the word is not found, the function should print: "Word not found in the file."
- The script should call this function after reading input from the user.

TASK 09:

- Create a program in C that sets up a shared memory segment.
- The child process should write a message (e.g., "Hello from child") into the shared memory.
- The parent process should read the message from the shared memory and display it.
- Ensure proper cleanup of the shared memory segment after communication.

TASK 10:

You are a system administrator tasked with preparing a report by creating and modifying text files on a Linux system.

- Create a new file named report.txt.
- Add the line "Operating Systems Lab - Mid Exam Practice" to the file.
- Append 5 more lines of text containing your name, date, and task description.
- Display the contents of the file on the terminal.
- Count and display the number of lines, words, and characters in the file.

THE END!